

David Cohen

M.Sc. Computer Science and Engineering student at the University of Bologna, four years of professional back-end experience, co-author of a peer-reviewed *Neural Networks* article.

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Cesena, Italy • Italian citizen (EU); no work permit required across the EU/EEA.

PUBLICATION

Abstractive summarization through the prism of decoding strategies

2026

G. Frisoni, L. Ragazzi, **D. Cohen**, G. Moro, A. Carbonaro, C. Sartori (*co-first authorship, equal contribution*)

Neural Networks 196, 108249, open access

- Contributed **Software, Investigation, Data curation, Formal analysis** and **Validation** (CRedit): built the experimental pipeline in Python (HuggingFace Transformers, PyTorch), ran the experiments on SLURM and processed the results.
- Compared decoding strategies for abstractive summarization across **five language models, six datasets and over 3,500 configurations**, scored with **ten automatic metrics** alongside human assessment, latency and carbon footprint.
- Released *Prism*, a public dataset of reference and model-generated summaries.

EDUCATION

M.Sc. Computer Science and Engineering

Sep 2022 – Expected Mar 2027

University of Bologna, Cesena, Italy

- Weighted average **28.5/30**, including *30 e lode* in Machine Learning and in Smart Vehicular Systems.

B.Sc. Computer Science and Engineering

Sep 2018 – Dec 2022

University of Bologna, Cesena, Italy

- Computer Engineering track. Thesis in natural language processing: refracting probability distributions in abstractive summarization by decoding strategies. Supervisors: Antonella Carbonaro and [Giacomo Frisoni](#).

EXPERIENCE

Teaching Tutor, Operating Systems

Sep 2025 – July 2026

University of Bologna, DISI Cesena, Italy

- Provisioned a local Moodle stack with Docker so the course could be tested offline.
- Designed the OFA question bank with graded difficulty, balanced taxonomy and structured categories.
- Docker, MariaDB, Moodle.

Full-Stack Developer

Jun 2021 – Jun 2023

Docenti.it, San Marino

- Developed the back end of a predictive engine for the national teacher rankings ([Aggiornamento Graduatorie](#)). It calculated scores under the ministerial guidelines and estimated next year's position with a statistical model built on historical rankings. It became the back-end engine of paid services such as **InCattedra**.
- Co-managed the company's CRM back end. Turned a prototype into the company's ticketing system, redesigned the check-out flow with the payment APIs and built the students' exam calendar.
- In my last year, developed the **IoMiLaureo** project and redesigned the REST APIs for the public website; helped migrate the internal CRM to an updated version, introduced shared team planning and **mentored my successors**.
- Java, Java EE, Hibernate, Spring MVC, JAX-RS, MariaDB, JSF, PrimeFaces, GWT; deployment on Nginx and Apache.

Full-Stack Developer Intern

Sep 2020 – Jun 2021

Docenti.it, San Marino

- Developed an algorithm that routes requests to tutors automatically, balancing the load by a weighted percentage; CRM modules, server-side development and bug fixing.
- Java EE (Hibernate, Spring MVC, JSF), MariaDB, GWT, JavaScript.

GC Group Srl, Rimini, Italy

- Configured and managed servers, workstations and network infrastructure; OS deployment, network monitoring and remote maintenance. On-site and remote IT support.
- Windows Server, Hyper-V, remote administration tooling.

SELECTED PROJECTS

Scala Robotics Simulator

team of 3, 5 sprints

Scala 3, cats-effect, gRPC, Q-learning, Deep Q-Networks, Python, Docker

- Built the **illumination engine** (occlusion rasterisation, pluggable field of view, bilinear interpolation, adaptive parallelism) and the **behaviour system**, a Kleisli-based DSL that composes partial rules into policies for obstacle avoidance, phototaxis, exploration and prioritised fallback.
- Owned the **phototaxis reinforcement-learning task** end to end: diagnosed a 43-million-state explosion that left Q-table coverage under 1%, redesigned the observation encoding down to **32 states** and reached a **90% success rate**.
- Trained a **Deep Q-Network** on 10x10 arenas that generalised to **86% / 95% / 100%** success on 5x5, 20x20 and 30x30 environments, after diagnosing three reward- and horizon-specification defects. Python agents train against the Scala engine over a **gRPC** contract; **65% measured coverage**, CI, Docker.

Code Metrics Analyzer

individual

Scala 3, Akka Typed, actor model, concurrency

- Built a parallel source-analysis application on an **Akka Typed** actor pipeline (crawler, work queue, N parallel processors, reducer, supervisor, UI bridge) that computes line-count metrics across Java repositories.
- Kept domain logic out of the actor layer behind a Strategy abstraction, so the metric changes without touching the topology; configurable parallelism, Cucumber BDD tests, CI.

Factory Manager, industrial IoT monitoring

team of 2

Node.js, MongoDB, MQTT, OpenAPI

- **Sole author of the Core REST service** inside a six-component system: domain model, HTTP API, authentication and authorization, persistence, tests and CI.
- Made telemetry ingestion **idempotent** so duplicate event identifiers return 409 Conflict and the coordinator can safely retry an at-least-once MQTT pipeline; invariants enforced by unique database indexes. **220 tests**; 1,280-line OpenAPI specification validated in CI.

Distributed Cooperative Pixel Art

individual

Kotlin, Java RMI, RabbitMQ, JVM 21

- Implemented the **same collaborative canvas twice**, first over centralised Java RMI and then over RabbitMQ fan-out, behind one shared network abstraction; both support join, snapshot, realtime updates, peer lifecycle handling and offline fallback.
- Compared the two models directly: the RMI version orders updates through a central sequencer, while the broker version trades that guarantee for decentralisation.

Autonomous Parking and Swarm Robotics

coursework

Python, CARLA, LiDAR, ARGoS, Lua, finite-state machines

- **S-Parking** (graded **30 e lode**, team of 2): six-state FSM that parks a car unassisted in CARLA from LiDAR and proximity sensing. Across a 24-run matrix of slot size, lateral offset and weather: **zero collisions**, and every infeasible slot refused rather than attempted.
- **Swarm aggregation** (individual): probabilistic FSM for **50 foot-bots** on range-and-bearing links of 10 bytes per message. Swept **81 parameter combinations**; simplifying the wander behaviour, not enriching it, was what made the swarm aggregate.

SKILLS

Programming languages: Scala 3, Java, Python, Kotlin, JavaScript and TypeScript, C++ (embedded, Arduino and ESP)**Research and ML:** Reinforcement learning (Q-learning, Deep Q-Networks), reward engineering, experimental design, natural language processing, PyTorch and HuggingFace, TensorFlow and Keras**Concurrency and distribution:** Actor model (Akka Typed), cats-effect, gRPC and Protobuf, Java RMI, MQTT, AMQP and RabbitMQ, microservices**Architecture:** Hexagonal architecture, CQRS, ports and adapters, domain-driven design; boundaries enforced in CI**Back end and data:** REST and OpenAPI, Spring MVC, Hibernate, Java EE, Node.js, MariaDB, MongoDB, Apache Spark on AWS EMR**Practices and tooling:** Docker, Git, GitHub Actions CI/CD, static analysis and coverage, BDD and TDD, Agile and Scrum**Spoken languages:** Italian (mother tongue), English (fluent), Greek (fluent spoken)**Certification:** Microsoft Server, GC Group Srl, Jun 2017